

Reconstructing the Earth from Geodesic Distances Using Classical MDS

Math 123 — Final Paper

Alex Bronstein
Owen Haber

Tufts University

July 20, 2026

Introduction

Imagine you wake up one day to find that every single map of the globe has been erased from existence. You, a clever and well-regarded mathematician, are given the daunting task of reassembling the maps of the globe from scratch in this dire time of need. However, all hope is not lost, along with your task, you receive the notebook of a very determined traveler who walked between thousands of cities all over the world and recorded their appropriate distances. With only these geodesic (great circle) distances could you really reconstruct an accurate map of the entire globe? Thankfully you took Math 123 at Tufts University, and remember a powerful technique that can be used to rebuild any space using just the distances between points, Multi-Dimensional Scaling.

Before we jump in to the math and its application, we need to cover the very important assumption that classical MDS operates under. MDS only guarantees a perfect reconstruction when the points that have recorded distances live in a flat, Euclidean space. This is due to the benefit of Euclidean squared distances forming a positive semi-definite (PSD) Gram matrix, which ensures that the distances are consistent with inner products in a flat space, and lets us know that a perfect reconstruction does exist.

However, in our situation, we aren't given the luxury of Euclidean distances. In this scenario we are instead only given the distances that our traveler walked, also known as the geodesic distances between cities. When working with geodesic distances (such as a distance traveled between two points on the surface of the Earth), we run into issues since the distances are between points along a manifold space (this manifold being a sphere). Specifically, the Gram matrix loses its property of being PSD, as negative eigenvalues may appear. In our case these negative eigenvalues indicate directions that simply don't exist in Euclidean space. As a result, global MDS reconstruction, both in 2D and 3D, displays significant distortion even after aligning them with their true coordinates via Procrustes analysis. The remainder of this project will be the analysis of this inherent error and a discussion of the different techniques we attempted to use to minimize it. .

Math Overview

Classical Multi-Dimensional Scaling (MDS) is a method that attempts to reconstruct the positions of points using only the pairwise distances between them. When these distances are from a flat Euclidean space, MDS is usually able to perfectly recover the point configuration up to a rotation, reflection, translation, and scaling. This is possible due to the fact that the squared Euclidean distance matrix contains enough information to reconstruct the Gram matrix of inner products, which encodes the geometry of the data. We can then use this Gram matrix to recover the general shape and relative positions of all of our points.

From Distances to an Embedding

Given a collection of points $x_1, \dots, x_n$ with pairwise distances d_{ij} , MDS begins by building the distance matrix D , whose entries are defined by

$$D_{ij} = d(x_i, x_j), \quad \text{the distance between points } x_i \text{ and } x_j$$

To turn these distances into inner products, MDS uses the centering matrix, J defined as

$$J = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top,$$

and then uses this to form the Gram matrix G :

$$G = -\frac{1}{2}JD^2J.$$

When the distances are Euclidean, G is PSD and also satisfies $G = XX^\top$, for some set of points $X \in \mathbb{R}^{n \times k}$. So if we can find G in terms of our distance matrix, we can use it to recover X . Therefore, we can attempt to factor G through its eigenvalue decomposition which would allow us to recover coordinates consistent with these inner products. To do this we begin by taking the eigenvalue decomposition of G :

$$G = V\Lambda V^\top,$$

we then can use the top k eigenvalues and eigenvectors to reconstruct our original coordinates:

$$X_{\text{mds}} = V_k \Lambda_k^{1/2}.$$

The rows of X_{mds} give the recovered point locations in k dimensions.

Procrustes Alignment

Even when MDS perfectly recovers the geometry of our data, the resulting configuration may be rotated, reflected, shifted, or scaled relative to the original coordinates. This is because with only pairwise distances, we know how points relate to one another (in other words we are able to derive their relative positions), but we have no way of recovering their absolute placement, orientation, or size. In order to recover this information, we can compare two shapes directly, applying *Procrustes alignment*, which finds the optimal transformation to best align the two point clouds, without changing any of their inherent shapes. Essentially we are aligning one point cloud X to another point cloud Y using only translations, rotations or stretches. Procrustes finds the scaling factor s , rotation matrix R , and translation vector t that minimize the Frobenius norm

$$\min_{s, R, t} \|Y - (sXR + t)\|_F.$$

Here:

- R is an orthogonal matrix representing a rotation (including a possible reflections),

- s rescales the reconstruction to match the target,
- t shifts every point in the cloud by the same amount.

In the context of MDS, Procrustes is crucial because it allows us to meaningfully compare our reconstruction to the ground truth coordinates without making any changes to the underlying structure of our embedding.

Global Reconstructions

For these reconstructions, we used the SimpleMaps world cities data set. After some basic cleaning, our data set contained the following: Each row represented one city. Along with each city was some basic information such as the country it belonged to, the city’s population, and most importantly the city’s coordinates. In total the data set contained roughly 50,000 cities.

Now that we understand our data and the math behind MDS, let’s actually explore our reconstructions. To begin, we wanted a baseline to understand just how much of an issue using geodesic distances instead of Euclidean ones would cause in our reconstruction. We began by taking the largest 2500 global cities to make computation and runtime more efficient for operations on the entire globe. After that, we prepared our data for MDS.

In most cases when dealing with points that lie on manifold space, there is no way to exactly compute distances between them. Fortunately for spherical manifolds, much like the globe, there exists an explicit formula that converts from coordinates to geodesic distance on a sphere with a set radius. This formula is called the haversine formula, which we used to generate our (great circle) distances between all of our selected cities.

$$d_{\text{hav}}(p, q) = 2R \arcsin \left(\sqrt{\sin^2 \left(\frac{\varphi_q - \varphi_p}{2} \right) + \cos(\varphi_p) \cos(\varphi_q) \sin^2 \left(\frac{\lambda_q - \lambda_p}{2} \right)} \right)$$

Where:

- φ = latitude (in *radians*),
- λ = longitude (in *radians*),
- R = Earth’s radius (we use $R \approx 6371$ km).

We then used this information to construct our distance matrix D , followed by our centering matrix, J and then used these to build our Gram matrix, G :

$$D_{ij} = d_{\text{hav}}(x_i, x_j), \quad J = I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top, \quad G = -\frac{1}{2} J D^2 J.$$

To finish off MDS, we performed an eigenvalue decomposition on our Gram matrix and took the top k positive eigenvalues (in this case $k = 3$ since we wanted a 3D reconstruction).

Lastly, we performed Procrustes alignment on our data to allow for easier visual comparisons of our reconstruction. For the 3D case, the ground truth matrix for Procrustes alignment was constructed by converting each city’s latitude and longitude into Earth-centered (ECEF) Cartesian coordinates using the known radius of the Earth. This would act as our consistent global reference frame for Procrustes alignment.

These were our results:

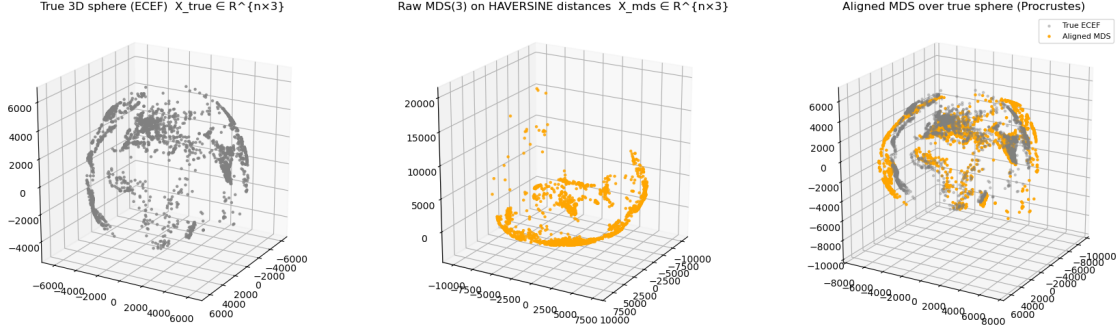
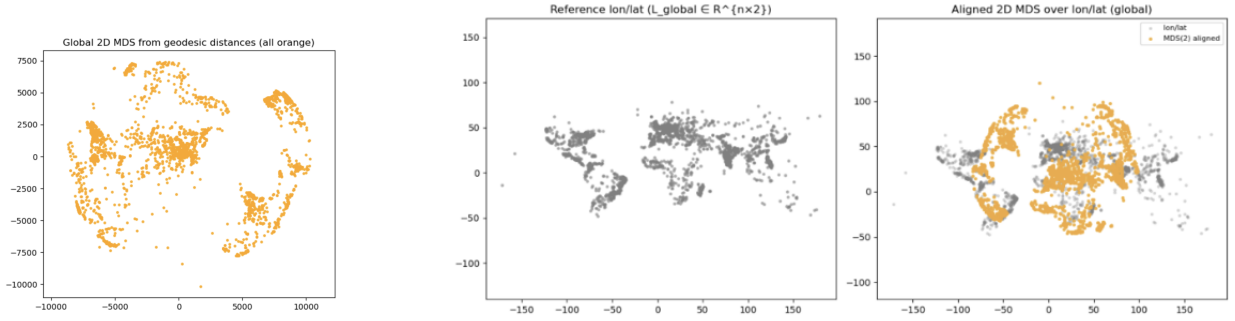


Figure 1: 3D ground truth, 3D global reconstruction before and after Procrustes

Figure 2: Avg Dist MAE = 802.42 KM

We then repeated this process for 2D (only using the top $k = 2$ eigenvectors) and again plotted our results both before and after applying Procrustes.



(a) 2D global reconstruction

(b) 2D ground truth alongside global reconstruction

Figure 3: Avg Dist MAE = 608.06 KM

Even though these reconstructions are clearly distorted, we can see that they are not just complete nonsense. Visually, we can see that these embeddings still preserve some broad structural relationships between regions. The key issue here is that geodesic distances do not satisfy the Euclidean identity $D_{ij}^2 = \|x_i - x_j\|^2$. As a result, we no longer guarantee that our Gram matrix is positive semi-definite. Once the PSD property is lost, it is impossible for MDS to produce an exact embedding and is instead forced to introduce distortions in order to account for inconsistencies in the distances.

Since we now have a grasp of how MDS handles these non-Euclidean distances, the natural next step is to attempt to find a way to force our distances to behave as similar as

possible to Euclidean ones.

A key line of intuition we follow to address this, is that even though the Earth is very curved globally, smaller, more local regions would likely behave approximately Euclidean. In other words, if we zoom in close enough, the geodesic distances become very similar to the Euclidean ones and can be treated as a more accurate approximation of them. When we apply MDS separately to continents (or even smaller regions), we would expect to see that our reconstructions improve drastically. We can then try stitching these pieces together using Procrustes in hopes that our results produce a more coherent and recognizable map of our globe.

Piecewise Reconstructions 2D

Since we wanted to see how local reconstructions compare to global ones, we began by partitioning our original dataset by continents and then sampled up to 900 cities per continent (if a continent contained less, we sampled all of them). We then repeated the MDS process, again using geodesic distances but, this time, we performed an entirely separate reconstruction for each individual continent.

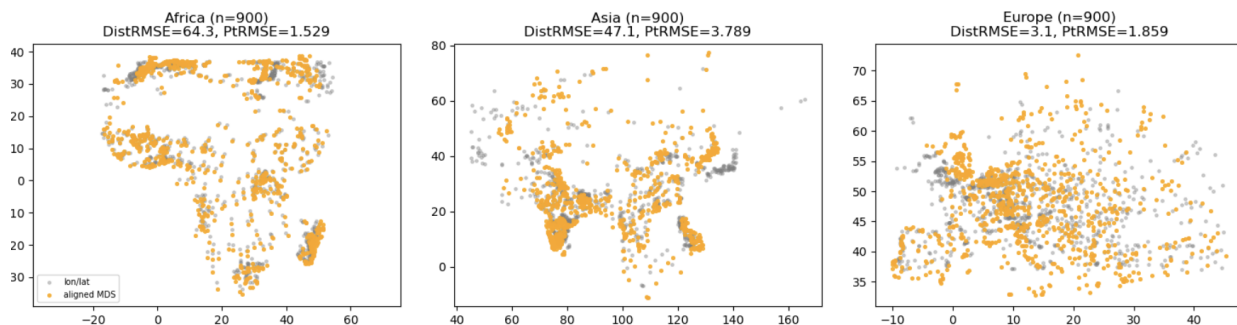


Figure 4: 2D reconstructions of Africa, Asia and Europe

These seem to be much more accurate than before; however, we can still see issues with scale and orientation. In particular, there is a visible amount of squashing and stretching happening to the geodesic distances when compared to their true 2D coordinates. This is due to the curvature of the Earth impacting our embeddings, and while noticeable, it is significantly less severe than the distortion from our global attempt at MDS.

Additionally, while MDS is able to capture structure, without a ground truth, there is no way to perfectly reconstruct our original scale and orientation. Luckily, dealing with these rotation and reflections is the exact problem that Procrustes is able to solve. Once we had all of our reconstructions, we used Procrustes alignment on each individual continent in order to piece all of them together into one map:

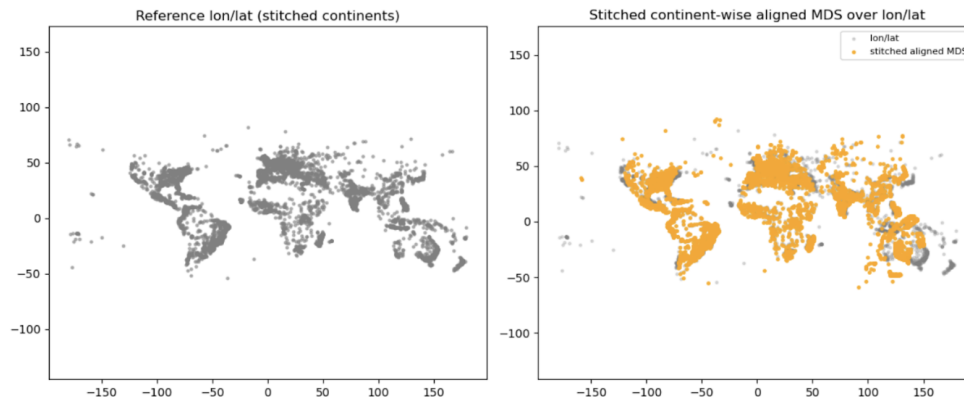


Figure 5: 2D reconstruction by continent

Figure 6: Avg Dist MAE = 66.19 KM

As we can see, these piecewise reconstructions are a significant improvement on our original global reconstructions. Something we noticed was that our 2D reconstruction showed some minor errors up to a reflection, even after Procrustes alignment. This could be due to the way classical MDS determines point configurations, only preserving shape and structure but unable to perfectly recover the original orientation. While Procrustes aims to fix this issue, it does not necessarily enforce orientation preservation. It may even select a reflected configuration if it minimizes the overall alignment error. In this case, for example, this is likely why even after applying Procrustes we can clearly see that Australia has been oriented incorrectly.

After seeing the significant improvements of performing piecewise reconstructions, our natural next step was to see if we could take this even further. What if instead of reconstructing continents we reconstructed individual countries? We proceeded by repeating very similar processes, performing individual reconstructions on countries instead of continents (with a max of 400 cities per country) and then again piecing them together to form a map using Procrustes alignment.

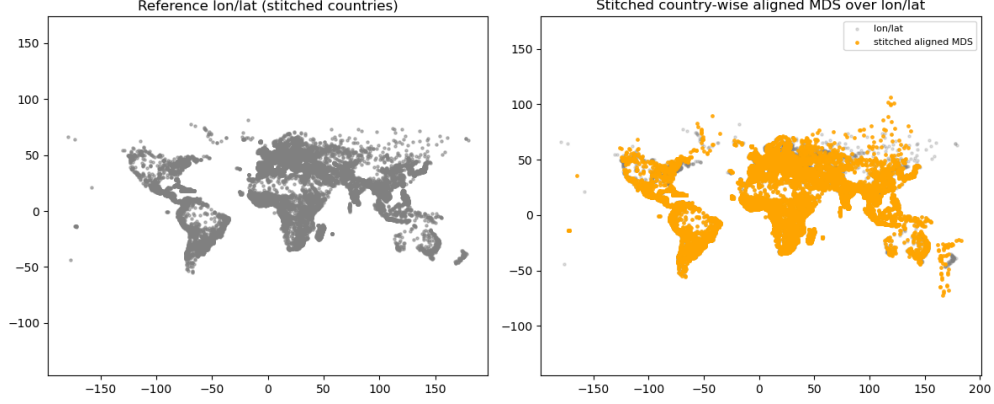


Figure 7: 2D reconstruction by country

Figure 8: Avg Dist MAE = 0.25 KM

These by country reconstructions showed significant improvements, especially in capturing local geometry and reducing distortions within regions.

Piecewise Reconstructions 3D

Next, we wanted to explore how these same techniques would translate to our 3D reconstructions. As before, we began by partitioning the dataset by continent and performed separate MDS reconstructions for each region, this time using $k = 3$ in order to generate 3D embeddings. We then applied Procrustes alignment to assemble the separately reconstructed continents into a unified 3D map.

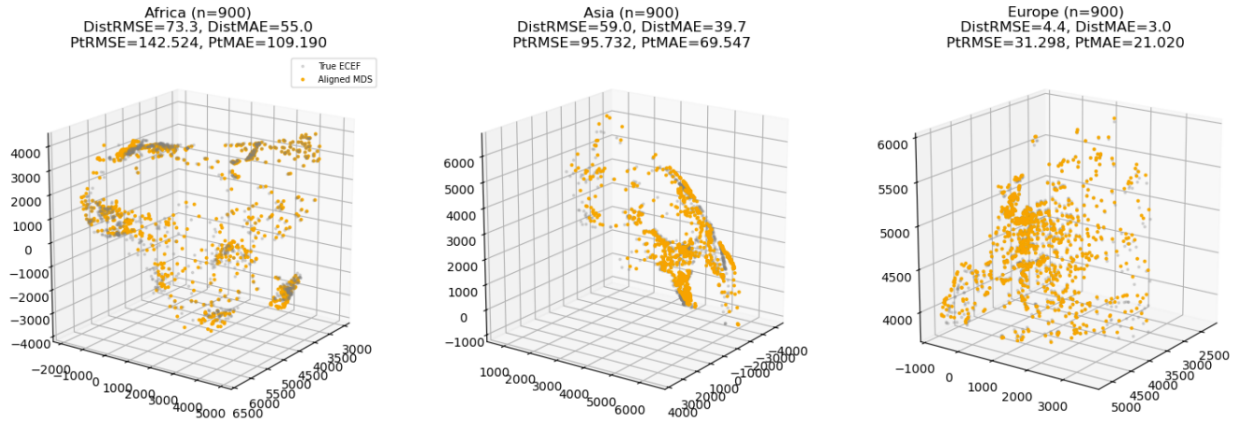


Figure 9: 3D reconstructions of Africa, Asia and Europe

Similarly to what we saw for 2D, the individual continent reconstructions appear to be quite accurate, with small amounts of visible distortion still prevalent. You can see the local curvature is much smaller as we zoom in, which matches our original intuition.

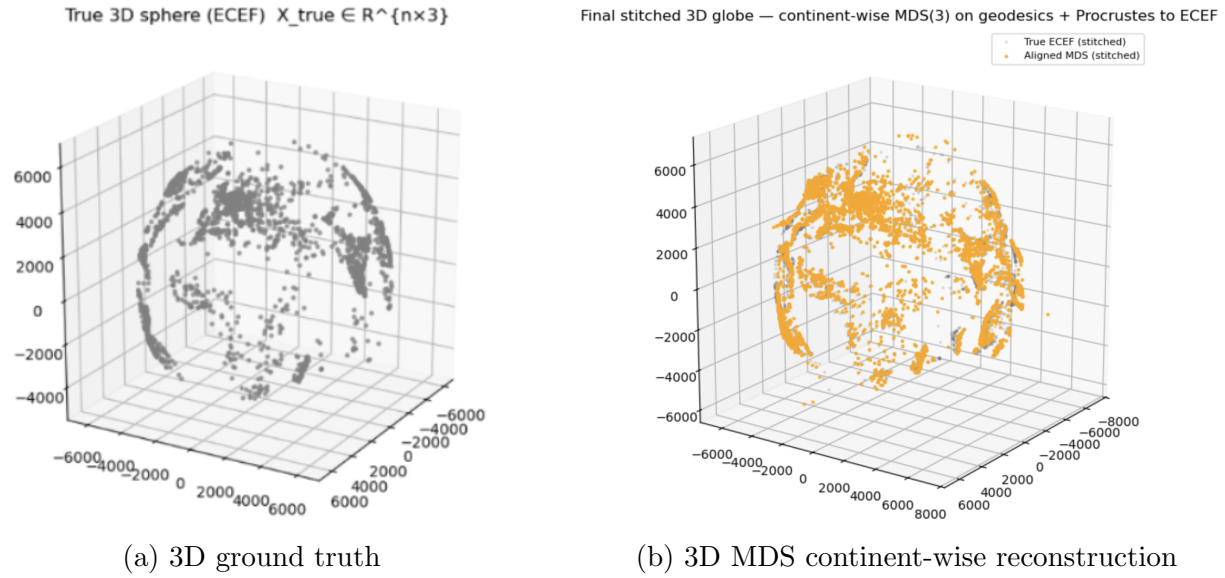


Figure 10: Avg Dist MAE = 60.87 KM

As expected, we again see significant improvements when compared to the original 3D global reconstruction. With this in mind, we proceeded to a per country based approach just like in 2D to see if we could further minimize our reconstruction error.

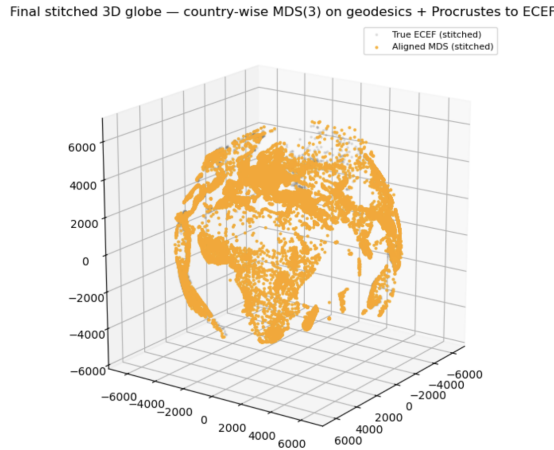


Figure 11: Avg Dist MAE = 0.29 KM

Although the increased locality noticeably improved accuracy compared to continent-wise and global reconstructions, we believed that this could be pushed even further. Encouraged by this clear trend, we explored whether zooming in even more would continue to improve reconstruction quality. To test this, we introduced a grid-based approach, splitting the globe into grids that span $5^\circ \times 5^\circ$ in latitude and longitude. We then applied MDS on each cell prior to alignment and stitching. Empty cells were not considered and in total we used 439 different sections out of 775, each containing as few as 10 and up to around 80 cities. Then, just as in the previous examples, we applied Procrustes individually on each grid in order to get a full reconstruction of the globe.

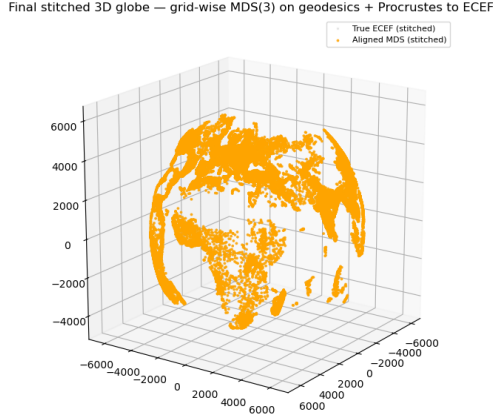


Figure 12: Avg Dist MAE = 0.01 KM

This resulted in our best reconstruction yet, achieving significantly lower reconstruction error than all of our previous attempts.

Error

Dim	Partition	NumParts	Avg Dist MAE	Avg Pt MAE	Total Pt MAE
2D	Global	1	608.06	20.30	20.30
2D	Continent	7	66.19	7.23	3.72
2D	Country	171	0.25	0.24	0.35

Table 1: Summary of partition error metrics

The results above fully support our hypothesis that localization is the driving factor for accuracy for this problem. From global, to continent, to country-based partitions, there is a very clear relationship between partition size of the manifold space and accuracy of MDS and Procrustes. This makes sense, since projecting an entire sphere into a flat space inevitably fails to preserve structure, whereas a thinner, more local slice (which becomes increasingly flat as we zoom in) can be represented more accurately.

Dim	Partition	NumParts	Avg Dist MAE	Avg Pt MAE	Total Pt MAE
3D	Global	1	802.42	421.82	421.82
3D	Continent	7	60.87	96.12	70.33
3D	Country	179	0.29	4.38	9.37
3D	Grid	439	0.01	0.41	0.41

Table 2: Summary of partition error metrics (3D)

We expected the error for the 3D reconstruction to be much better than 2D, where we lose a dimension of information, but our results largely supported the opposite. We can observe the same trend in error when we increase the number of partitions, but at some of the

partition sizes shared with 2D, the 2D reconstructions perform better. This is likely because embedding in three dimensions allows curvature-induced distortion from the geodesic distances to manifest more freely, whereas 2D constrains and nullifies some of these distortions. We can observe this effect in the 3D reconstructions, where reconstructed points appear to be drifting off the spherical surface of the globe.

Between both 2D and 3D problems, the reconstruction error decreases monotonically as partitions become more localized, which confirms that MDS has the best performance when geodesic distances are restricted to approximately Euclidean regions. Ultimately, the finest partitions, being our $5^\circ \times 5^\circ$ grids, achieve the lowest errors by minimizing curvature effects.

Summary & Take Aways

Overall, classical MDS performed surprisingly well considering that the underlying space was non-Euclidean, which broke the fundamental rules for the operation’s problem setting. While it performed poorly at the beginning, its later success relied on heavily restricting the process to local regions. The fact that the error was monotonic as we increased localization suggests that this problem setting was a best case scenario for running MDS on a curved manifold. Other manifold spaces don’t guarantee an explicit way to compute geodesic distances, and creating a ground truth in Euclidean space like we did for this project can often be impossible. Additionally, if the Earth lay on a more complex or irregular manifold, these partitioning strategies would likely have failed to recover an interpretable global structure.

Considering the limits of our approach for this problem, increasing the number of data points (cities) and decreasing the partition size would continue the monotonic improvement of the reconstruction error. If we had funding to purchase SimpleMaps’ larger data set with 4 million cities, and had access to a cluster to run MDS in parallel, we could explore these limits and see how close to 0 MSE error smaller partitions and denser data could yield. However, this could at some point lead to a loss of global structure as partitions become too localized, making the final construction rely too heavily on local alignments. This problem has shown that improving accuracy requires shrinking partitions until curvature is negligible, but doing so replaces one global reconstruction problem with many local ones that must be stitched together. This reminds us that MDS can only avoid curvature, never model it, even as we stretch our technique to the limit.

The findings of this problem motivate the need for data techniques that are aware of, or account for, manifold structure. The natural extension of this problem, or next step, would be to explore methods equipped for this problem, such as ISOMAP. ISOMAP in contrast to MDS constructs a neighborhood graph that preserves local Euclidean relationships and then approximates geodesic distances through the shortest paths, a more directed and universally applicable localization approach for manifold approximation. While our results show that classical MDS can approximate curved geometry through localization, true reconstruction of manifold spaces requires methods that explicitly model the manifold itself.

References

- [1] SimpleMaps. *World Cities Database*. Available at: <https://simplemaps.com/data/world-cities>, accessed October 18, 2025.